

UNDERSTANDING HACKING AND CYBERSECURITY

An educational overview of hacking concepts, types, and defense strategies

WHAT IS HACKING?

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CORE IDEA

Definition: Unauthorized access to computer systems or networks



- 02  Motivations range from curiosity to financial gain or activism
- 03  Can involve exploiting software vulnerabilities or human psychology
- 04  Legal consequences vary by jurisdiction and intent

Hacking is unauthorized system access with varied motivations and legal implications.

TYPES OF HACKERS

01

White Hat

Ethical hackers who test systems with permission

02

Black Hat

Malicious actors who breach systems illegally

03

Grey Hat

Operate without permission but may report vulnerabilities

04



Script Kiddies

Inexperienced users relying on pre-made tools

Hacker classification depends on intent, authorization, and ethical stance.

COMMON ATTACK VECTORS

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- 01 Phishing: Deceptive emails or messages stealing credentials
- 02 {...} Malware: Malicious software like ransomware, trojans, or spyware

- 03  SQL Injection: Exploiting database vulnerabilities via input fields
- 04  Social Engineering: Manipulating people to bypass security controls

Attacks target both technical vulnerabilities and human psychology.

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DEFENSIVE CYBERSECURITY MEASURES

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01



Firewalls and intrusion detection
systems monitor network traffic

02



Regular patching closes known
software vulnerabilities

03



Multi-factor authentication adds
verification beyond passwords

04



Security awareness training
reduces human error risks

Layered defenses combining technology and education provide the strongest protection.

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ETHICAL HACKING AND PENETRATION TESTING

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CORE IDEA

Authorized simulated attacks identify weaknesses before criminals do



- 02 Follows structured methodologies like OWASP or PTES
- 03  Reports include risk ratings and remediation recommendations
- 04 Certifications include CEH, OSCP, and GPEN

Ethical hacking proactively strengthens security through controlled testing.

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KEY TAKEAWAYS

UNDERSTANDING HACKING AND CYBERSECURITY

01



Hacking spans a spectrum from criminal activity to legitimate security testing

02



Understanding attack methods enables better defense strategies

03

Human factors remain a critical vulnerability in any system

04



Continuous learning and adaptation are essential in cybersecurity

Cybersecurity requires understanding threats, layered defenses, and ongoing vigilance.

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THANK YOU

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